Practical no. 14

Aim - Design a web dashboard using Flask. Control relay module via web.

Hardware Requirements

1. Raspberry Pi Model A/B/B+
2. Jumper wires (Female to Male)
3. Relay Module

- **Software Requirements**
  1. Raspbian Stretch OS

1. **Connect your Relay Module with Raspberry Pi's GPIO Pins.**

Relay	Relay Input	RPI Physical Pin	Raspberry Function
VCC	POWER	2	VCC
GND	GND	6	GND
CH 1	IN 1	12	GPIO 18

- **Code:-**
- **1. Python code:**

```
from flask import Flask, render_template_string
import RPi.GPIO as GPIO
app = Flask(__name__)
RELAY = 17
# GPIO setup
GPIO.setmode(GPIO.BCM)
GPIO.setup(RELAY, GPIO.OUT)
GPIO.output(RELAY, GPIO.HIGH)
# HTML page
HTML = """
<!DOCTYPE html>
<html>
<head>
    <title>Relay Control</title>
    <meta name="viewport" content="width=device-width, initial-scale=1">
    <style>
        body {
            font-family: Arial;
            text-align: center;
            margin-top: 70px;
        }
        .status {
            font-size: 28px;
            font-weight: bold;
        }
        button {
            font-size: 25px;
            padding: 20px 40px;
            margin: 15px;
            color: white;
            border: none;
            border-radius: 8px;
        }
```

```
        .on {
            background: green;
        }
        .off {
            background: red;
        }
    </style>
</head>
<body>
    <h1>Raspberry Pi Relay</h1>
    {% if state %}
        <div class="status" style="color:green;">
            Status: ON
        </div>
    {% else %}
        <div class="status" style="color:red;">
            Status: OFF
        </div>
    {% endif %}
    <a href="/on">
        <button class="on">ON</button>
    </a>
    <a href="/off">
        <button class="off">OFF</button>
    </a>
</body>
</html>
"""

@app.route("/")
def home():
    state = GPIO.input(RELAY) == GPIO.LOW
    return render_template_string(HTML, state=state)
@app.route("/on")
def turn_on():
    GPIO.output(RELAY, GPIO.LOW)
```

```
    return render_template_string(HTML, state=True)
@app.route("/off")
def turn_off():
    GPIO.output(RELAY, GPIO.HIGH)
    return render_template_string(HTML, state=False)
try:
    app.run(host="0.0.0.0", port=5000)
finally:
    GPIO.cleanup()
```

1. Open the terminal

1. *sudo apt update*
2. *sudo apt install python3-flask*
3. *python3 –c “import flask ; print (‘Flask OK’)”*
4. *nano relay.py*
5. *python3 relay.py*

2. Open another terminal

Write a command (hostname -I)

